

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

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Subject: Java

Assignment No.: 6

Title: Movie Ticket Booking System

Date: 03/09/2025

Java
Assignment - 6

Name: Riddhee Kulkarni

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Aim:

To demonstrate the creation and execution of multiple threads using both the Thread class and the Runnable interface in the context of a Movie ticket booking system.

Theory:

1. Threading

A thread in Java is a lightweight unit of a process that allows a program to perform multiple tasks at the same time. It represents a single sequence of execution within a program.

2. Multithreading

Multithreading is the process of executing two or more threads simultaneously to maximize CPU utilization. It allows concurrent execution of tasks, making programs faster and more responsive.

3. Runnable interface

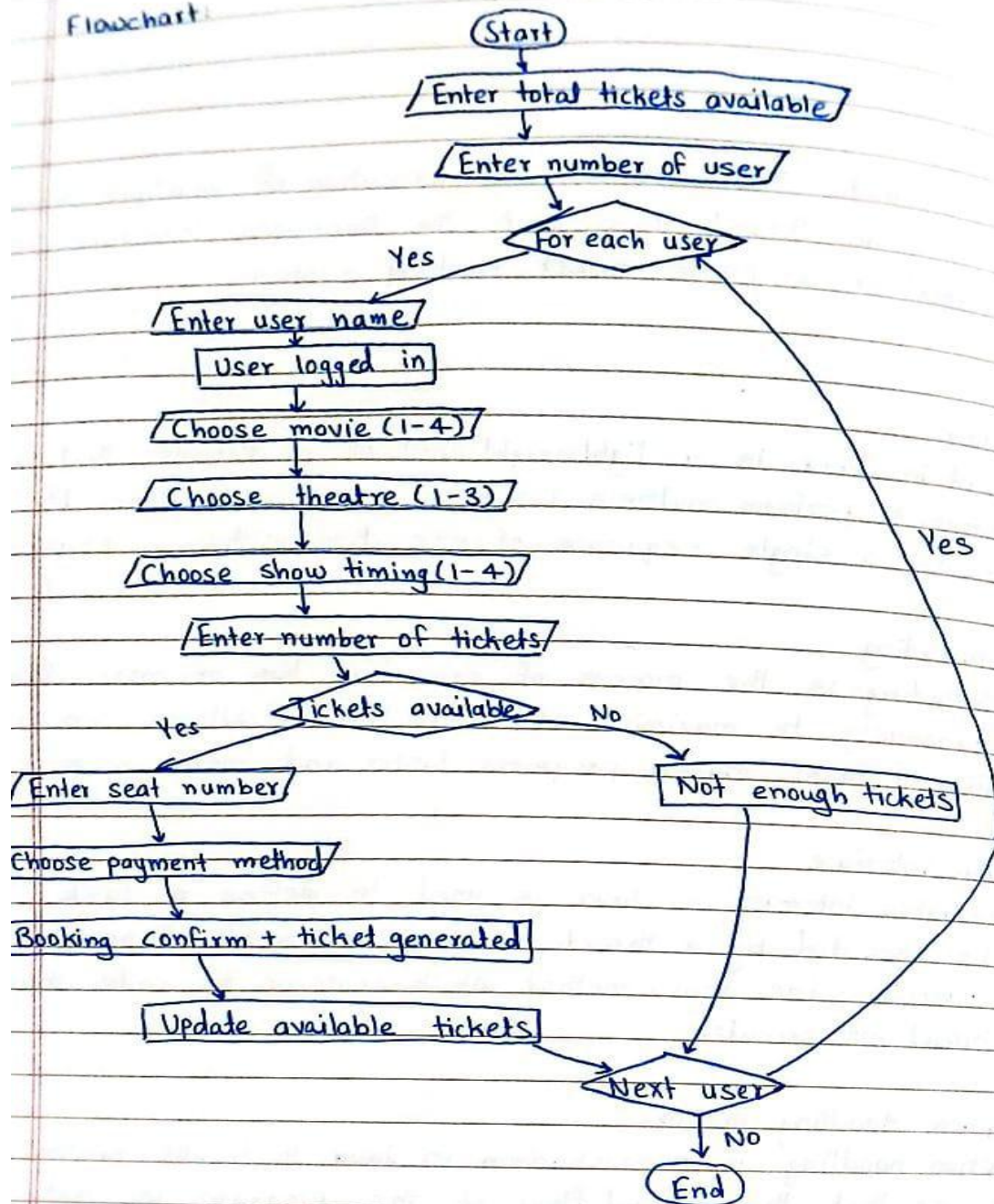
The runnable interface in Java is used to define a task that can be executed by a thread. A class implements Runnable and overrides the run() method, which contains the code that the thread will execute.

4. Exception Handling ~~is~~

Exception handling is a mechanism in Java to handle runtime errors so that the normal flow of the program is not disrupted. It uses keywords like try, catch, throw, throws.

and finally to gracefully manage errors.

Flowchart:



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Code:

```
package Assignment6;
import java.util.*;

class TicketBooking
{
    private int availableTickets;

    public TicketBooking(int totalTickets)
    {
        this.availableTickets = totalTickets;
    }

    public synchronized void bookTicket(String user, Scanner sc)
    {
        try
        {
            System.out.println("\n" + user + " → Logged in");

            System.out.println("Available movies:");
            System.out.println("1. Pathaan");
            System.out.println("2. Gadar 2");
            System.out.println("3. 3 Idiots");
            System.out.println("4. Dangal");
            System.out.print("Choose a movie (1-4): ");
            int movieChoice = Integer.parseInt(sc.nextLine());

            String movie;
            switch (movieChoice)
            {
                case 1: movie = "Pathaan"; break;
                case 2: movie = "Gadar 2"; break;
                case 3: movie = "3 Idiots"; break;
                case 4: movie = "Dangal"; break;
                default: movie = "Unknown"; break;
            }

            System.out.println("Available theatres:");
            System.out.println("1. PVR");
            System.out.println("2. INOX");
            System.out.println("3. Cinepolis");
            System.out.print("Choose a theatre (1-3): ");
            int theatreChoice = Integer.parseInt(sc.nextLine());

            String theatre;
            switch (theatreChoice)
            {
                case 1: theatre = "PVR"; break;
                case 2: theatre = "INOX"; break;
                case 3: theatre = "Cinepolis"; break;
                default: theatre = "Unknown"; break;
            }

            System.out.println("Available show timings:");
            System.out.println("1. Morning 10 AM");
            System.out.println("2. Afternoon 2 PM");
            System.out.println("3. Evening 6 PM");
            System.out.println("4. Night 9 PM");
            System.out.print("Choose a show (1-4): ");
            int showChoice = Integer.parseInt(sc.nextLine());

            String show;
            switch (showChoice)
            {
                case 1: show = "Morning 10 AM"; break;
                case 2: show = "Afternoon 2 PM"; break;
                case 3: show = "Evening 6 PM"; break;
                case 4: show = "Night 9 PM"; break;
                default: show = "Unknown"; break;
            }
        }
    }
}
```

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```
System.out.print("How many tickets do you want? ");
int tickets = Integer.parseInt(sc.nextLine());

if (tickets <= availableTickets)
{
    List<String> seats = new ArrayList<>();
    for (int i = 1; i <= tickets; i++)
    {
        System.out.print("Enter seat number " + i + ": ");
        seats.add(sc.nextLine());
    }

    System.out.println("Payment methods:");
    System.out.println("1. UPI");
    System.out.println("2. Card");
    System.out.println("3. Cash");
    System.out.print("Choose payment method (1-3): ");
    int payChoice = Integer.parseInt(sc.nextLine());

    String payment;
    switch (payChoice)
    {
        case 1: payment = "UPI"; break;
        case 2: payment = "Card"; break;
        case 3: payment = "Cash"; break;
        default: payment = "Unknown"; break;
    }

    availableTickets -= tickets;
    System.out.println("\nBooking confirmed for " + movie + " at " + theatre + " (" + show + ")");
    System.out.println("Seats: " + seats);
    System.out.println("Payment method: " + payment);
    System.out.println("Tickets left: " + availableTickets);

    System.out.println("Ticket generated for " + user + "\n");
}
else
{
    System.out.println("Sorry " + user + ", not enough tickets available.");
}
}
catch (Exception e)
{
    System.out.println("Error for " + user + ": " + e.getMessage());
}
}

class UserThread extends Thread
{
    private TicketBooking booking;
    private String user;
    private Scanner sc;

    public UserThread(TicketBooking booking, String user, Scanner sc)
    {
        this.booking = booking;
        this.user = user;
        this.sc = sc;
    }

    public void run()
    {
        booking.bookTicket(user, sc);
    }
}

class UserRunnable implements Runnable
```

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```
private TicketBooking booking;
private String user;
private Scanner sc;

public UserRunnable(TicketBooking booking, String user, Scanner sc)
{
    this.booking = booking;
    this.user = user;
    this.sc = sc;
}

public void run()
{
    booking.bookTicket(user, sc);
}
}

public class MovieTicketBookingSystem
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter total tickets available: ");
        int totalTickets = Integer.parseInt(sc.nextLine());

        TicketBooking booking = new TicketBooking(totalTickets);

        System.out.print("Enter number of users: ");
        int n = Integer.parseInt(sc.nextLine());

        for (int i = 1; i <= n; i++)
        {
            System.out.print("Enter name of user " + i + ": ");

            String name = sc.nextLine();

            Thread t;
            if (i % 2 == 0)
            {
                t = new UserThread(booking, name, sc);
            }
            else
            {
                t = new Thread(new UserRunnable(booking, name, sc));
            }
            t.start();
            try
            {
                t.join();
            }
            catch (InterruptedException e)
            {
                System.out.println("Thread interrupted: " + e.getMessage());
            }
        }

        sc.close();
    }
}
```

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Output:

```
Enter total tickets available: 5
Enter number of users: 3
Enter name of user 1: Riddhee
```

```
Riddhee → Logged in
Available movies:
1. Pathaan
2. Gadar 2
3. 3 Idiots
4. Dangal
Choose a movie (1-4): 3
Available theatres:
1. PVR
2. INOX
3. Cinepolis
Choose a theatre (1-3): 1
Available show timings:
1. Morning 10 AM
2. Afternoon 2 PM
3. Evening 6 PM
4. Night 9 PM
Choose a show (1-4): 3
How many tickets do you want? 3
Enter seat number 1: 3
Enter seat number 2: 4
Enter seat number 3: 5
Payment methods:
1. UPI
2. Card
3. Cash
Choose payment method (1-3): 1
```

```
Booking confirmed for 3 Idiots at PVR (Evening 6 PM)
```

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Seats: [3, 4, 5]
Payment method: UPI
Tickets left: 2
Ticket generated for Riddhee

Enter name of user 2: Rujuta

Rujuta → Logged in

Available movies:

1. Pathaan
2. Gadar 2
3. 3 Idiots
4. Dangal

Choose a movie (1-4): 4

Available theatres:

1. PVR
2. INOX
3. Cinepolis

Choose a theatre (1-3): 3

Available show timings:

1. Morning 10 AM
2. Afternoon 2 PM
3. Evening 6 PM
4. Night 9 PM

Choose a show (1-4): 2

How many tickets do you want? 2

Enter seat number 1: 45

Enter seat number 2: 46

Payment methods:

1. UPI
2. Card
3. Cash

Choose payment method (1-3): 3

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Booking confirmed for Dangal at Cinepolis (Afternoon 2 PM)
Seats: [45, 46]
Payment method: Cash
Tickets left: 0
Ticket generated for Rujuta

Enter name of user 3: Nandini

Nandini → Logged in
Available movies:
1. Pathaan
2. Gadar 2
3. 3 Idiots
4. Dangal
Choose a movie (1-4): 2
Available theatres:
1. PVR
2. INOX
3. Cinepolis
Choose a theatre (1-3): 2
Available show timings:
1. Morning 10 AM
2. Afternoon 2 PM
3. Evening 6 PM
4. Night 9 PM
Choose a show (1-4): 2
How many tickets do you want? 3
Sorry Nandini, not enough tickets available.

Conclusion:

This program successfully demonstrates the use of multi-threading in Java by creating and executing threads using both the Thread class and Runnable interface.